Problem Set 2 Solution

Part One: Computer exercises

1. Assuming each flip is independent, simulate `1,000` fair coin flips and plot the cumulative fraction of heads against the number of flips (n=1, 2,...,1000). Include a horizontal reference line at `0.5`. In your answer, clearly specify the exact steps of your simulation process. Finally, discuss your findings with reference to the Law of Large Numbers

```
# it's optional to set a seed number
set.seed(2026)

# simulate 1,000 coin flips by denoting head if value above 0.5, tail othe
rwise
coin_flips <- runif(1000) > 0.5    # TRUE TRUE FALSE FALSE TRUE FALSE

# calculate cumulative number and fraction of heads for coin flips of 1,
2, 3, ..., 1000
cum_heads <- cumsum(coin_flips)
frac_heads <- cum_heads / 1:1000

# record the result of each flip in dataframe
df <- data.frame(
  flips = 1:1000,
  coin_outcome = coin_flips,
  cumulative_heads = cum_heads,
  fraction_heads = frac_heads
)

# take a look at the fitst few records
head(df)

# output

  flips coin_outcome cumulative_heads fraction_heads
1      1           TRUE                1      1.0000000
2      2           TRUE                2      1.0000000
3      3          FALSE                2      0.6666667
4      4          FALSE                2      0.5000000
5      5           TRUE                3      0.6000000
6      6          FALSE                3      0.5000000
```

```r
library(ggplot2)

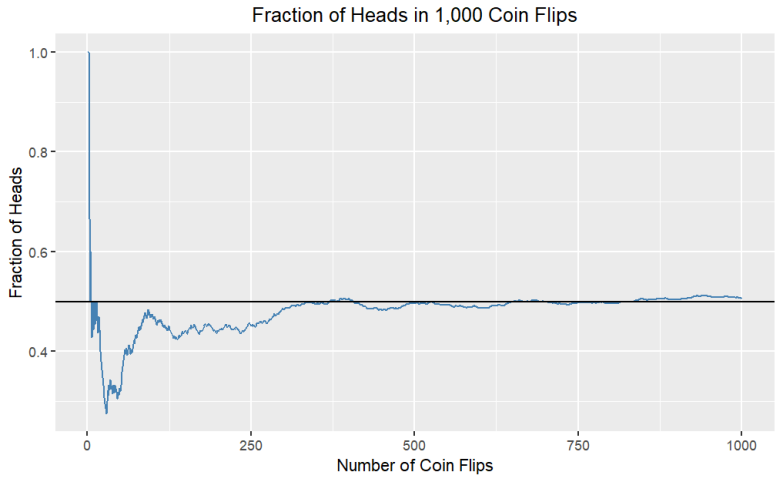
# draw the fraction of head over each flip
ggplot(df, aes(x = flips, y = fraction_heads)) +

  ## draw the fraction of head
  geom_line(color = "steelblue") +

  ## baseline fraction is 0.5
  geom_hline(aes(yintercept = 0.5), color = "black") +

  ## add label to make the graph more readable
  labs(
    title = "Fraction of Heads in 1,000 Coin Flips",
    x = "Number of Coin Flips",
    y = "Fraction of Heads"
  ) +

  ## center the graph title
  theme(plot.title = element_text(hjust = 0.5))
```



2. Draw a single random sample of `10,000` values from a normal distribution with μ = `2` and σ = `2.5`. Using this sample, plot four separate histograms using a bin width of 1, 0.1, 0.01, and 0.001. Discuss your findings by comparing the four plots, noting how the visual representation of the underlying distribution changes as the bin width becomes extremely small

```r
set.seed(2026)

# Draw a random sample of 10,000 values
sample <- rnorm(n = 10000, mean = 2, sd = 2.5)
```

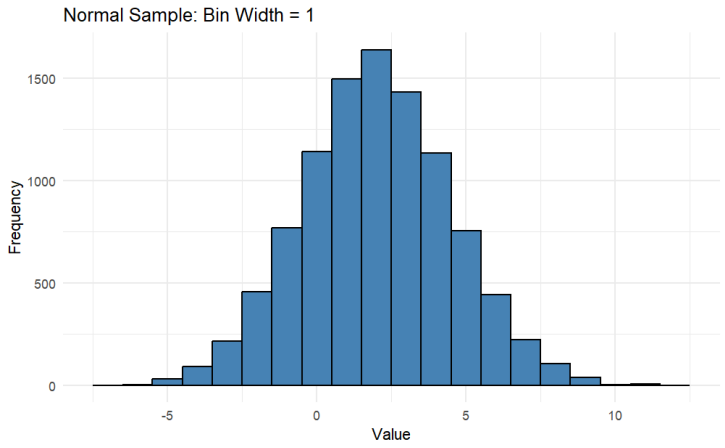
```r
# Create a data frame
data <- data.frame(x = sample)

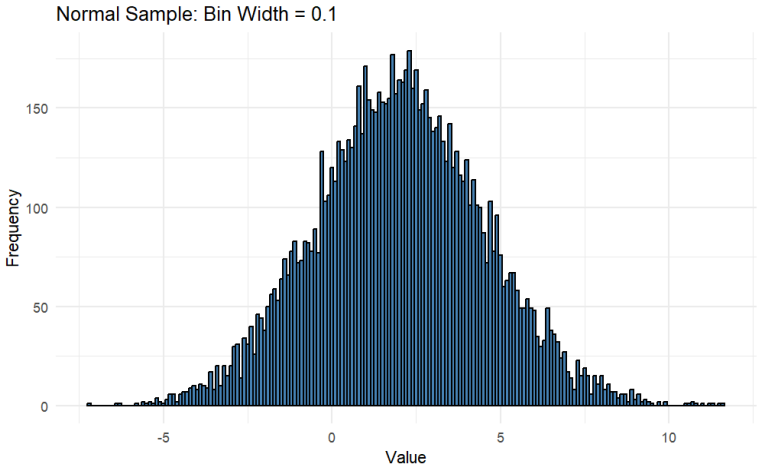
# Histogram with bin width = 1
ggplot(data, aes(x = x)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "black") +
  labs(
    title = "Normal Sample: Bin Width = 1",
    x = "Value",
    y = "Frequency"
  ) +
  theme_minimal()

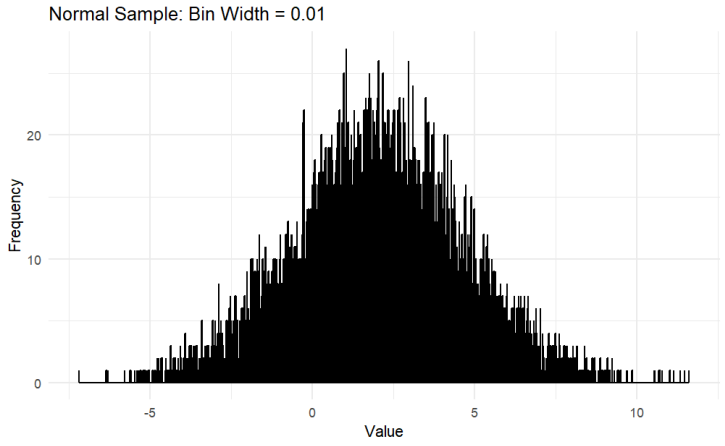
# Histogram with bin width = 0.1
ggplot(data, aes(x = x)) +
  geom_histogram(binwidth = 0.1, fill = "steelblue", color = "black") +
  labs(
    title = "Normal Sample: Bin Width = 0.1",
    x = "Value",
    y = "Frequency"
  ) +
  theme_minimal()

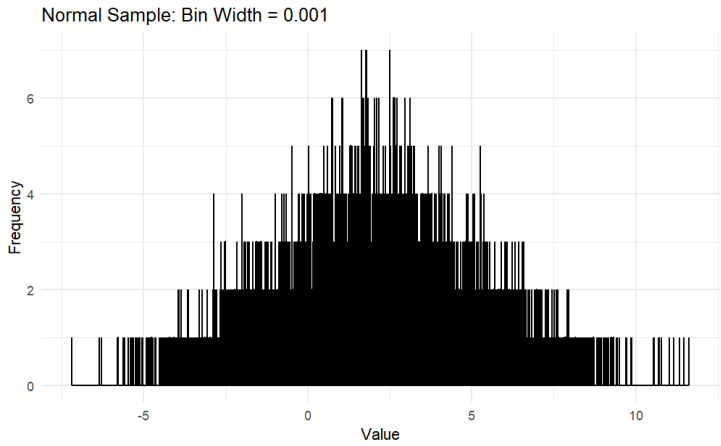
# Histogram with bin width = 0.01
ggplot(data, aes(x = x)) +
  geom_histogram(binwidth = 0.01, fill = "steelblue", color = "black") +
  labs(
    title = "Normal Sample: Bin Width = 0.01",
    x = "Value",
    y = "Frequency"
  ) +
  theme_minimal()

# Histogram with bin width = 0.001
ggplot(data, aes(x = x)) +
  geom_histogram(binwidth = 0.001, fill = "steelblue", color = "black") +
  labs(
    title = "Normal Sample: Bin Width = 0.001",
    x = "Value",
    y = "Frequency"
  ) +
  theme_minimal()
```


Normal Sample: Bin Width = 1
Frequency
1500
1000
500
0
-5
0
5
10
Value


Normal Sample: Bin Width = 0.1
Frequency
150
100
50
0
-5
0
5
10
Value


Normal Sample: Bin Width = 0.01
Frequency
20
10
0
-5
0
5
10
Value



3. Draw `200` values from a binomial distribution with p = `0.35` and n = `10`, `50`, `100`, and `500`. Note that n here is the number of trials for a binomial distribution. Make a histogram for each case. What do you find as n gets larger?

```
set.seed(2026)

# define distribution parameter and trial size
p <- 0.35
sample_num <- 200

# draw the random binomial samples according to different sample size
draw_size10   <- rbinom(sample_num, size = 10,  prob = p)
draw_size50  <- rbinom(sample_num, size = 50,  prob = p)
draw_size100  <- rbinom(sample_num, size = 100, prob = p)
draw_size500 <- rbinom(sample_num, size = 500, prob = p)

# compute the simulated mean and standard deviation according to different
draw sizes
sim_result <- data.frame(
  sample_size = c(10, 50, 100, 500),
  mean = c(mean(draw_size10), mean(draw_size50), mean(draw_size100), mean
(draw_size500)),
  sd = c(sd(draw_size10), sd(draw_size50), sd(draw_size100), sd(draw_size5
00))
)
print(sim_result)
```

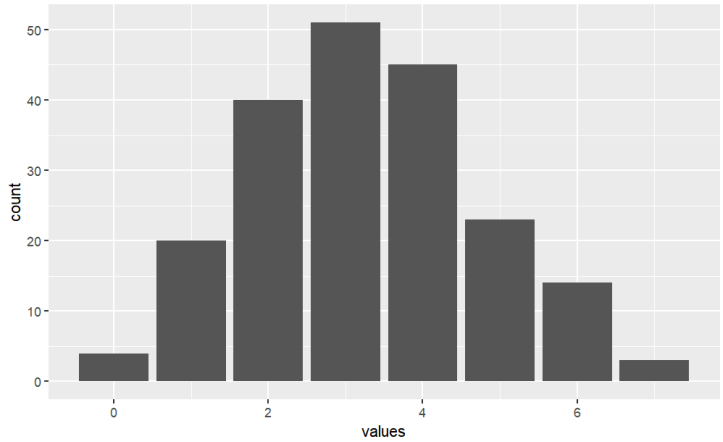
```
# the histogram when draw size = 10
df_size10 <- data.frame(
  values = draw_size10
)
```

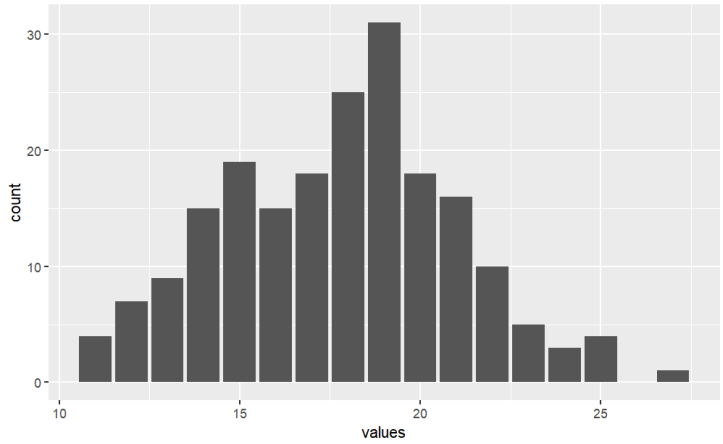
```r
ggplot(df_size10, aes(x = values)) +
  geom_bar()

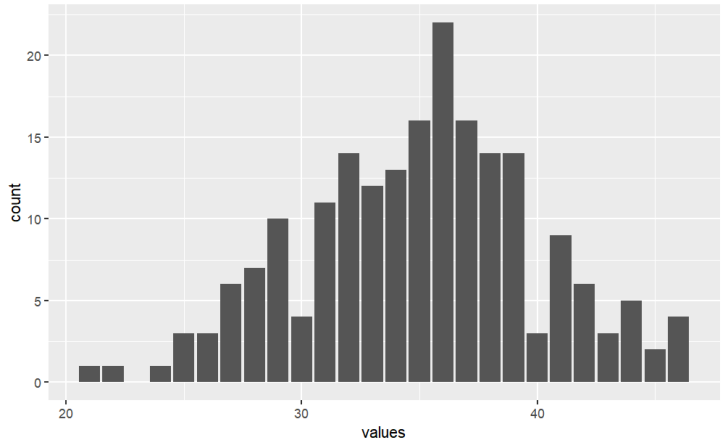
# the histogram when draw size = 50
df_size50 <- data.frame(
  values = draw_size50
)
ggplot(df_size50, aes(x = values)) +
  geom_bar()

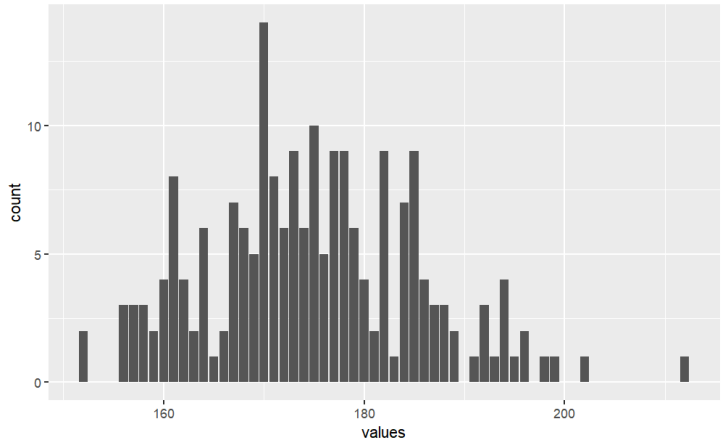
# the histogram when draw size = 100
df_size100 <- data.frame(
  values = draw_size100
)
ggplot(df_size100, aes(x = values)) +
  geom_bar()

# the histogram when draw size = 500
df_size500 <- data.frame(
  values = draw_size500
)
ggplot(df_size500, aes(x = values)) +
  geom_bar()
```









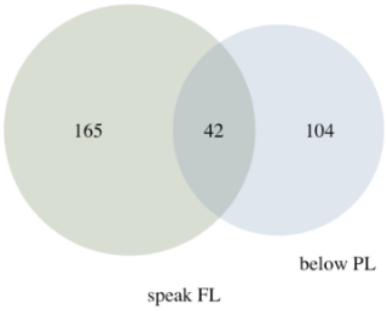
Part Two: Textbook exercises

Question 3.8

We denote **Poverty Line** as **PL**, and **Foreign Language** as **FL**.

a. No, there are people who are both living below the poverty line and speak a language other than English at home.

b. The Venn diagram is shown below.



c. Each person living below the poverty line either speaks only English at home or doesn't. Since 14.6% of Americans live below the poverty line and 4.2% speak a language other than English at home, the other 10.4% only speak English at home.

d. Using the **General Addition Rule**:

$$\begin{aligned} P(\text{below PL or speak FL}) &= P(\text{below PL}) + P(\text{FL}) - P(\text{both}) \\ &= 0.146 + 0.207 - 0.042 \\ &= 0.311 \end{aligned}$$

e. Probability of **neither** below PL **nor** speak FL:

$$\begin{aligned} P(\text{neither below PL nor speak FL}) &= 1 - P(\text{below PL or speak FL}) \\ &= 1 - 0.311 \\ &= 0.689 \end{aligned}$$

f. Two approaches to assess independence:

1. **Multiplication rule (if independent)**

$$P(\text{below PL}) \times P(FL) = 0.146 \times 0.207 = 0.030$$

This does **not** equal $P(\text{below PL and speak FL}) = 0.042$, therefore the events are **dependent**.

2. **Bayes' theorem**

If the two events were independent, then

$$P(\text{below PL} \mid \text{FL}) = P(\text{below PL})$$

Using Bayes' theorem:

$$P(\text{below PL} \mid \text{FL}) = \frac{P(\text{below PL and speak FL})}{P(FL))} = \frac{0.042}{0.207} \approx 0.203$$

Since this probability is different from $P(\text{below PL}) = 0.146$, we determine that the two events are **dependent**.

Question 3.16

a. No, there are individuals who are both in excellent health and **have** health coverage.

b. Overall probability: $P(\text{excellent health}) = 0.2329$

c. Conditional on **having** health coverage:

$$P(\text{excellent health} \mid \text{coverage}) = \frac{0.2099}{0.8738} \approx 0.24$$

d. Conditional on **not having** health coverage:

$$P(\text{excellent health} \mid \text{no coverage}) = \frac{0.0230}{0.1262} \approx 0.18$$

e. Conclusion: Since the probability of excellent health differs between coverage groups (0.24 vs 0.18), the variables **are not independent**.

Question 3.34

a. The probability model and the calculation of average revenue per passenger (expected value)

Event	X	$P(X)$	$X \cdot P(X)$	$[X - E[X]]^2$	$[X - E[X]]^2 \cdot P(X)$
No baggage	0	0.54	0	246.49	133.10
1 checked bag	25	0.34	8.5	86.49	29.41
2 checked bags	60	0.12	7.2	1962.49	235.50

1. expected value:

$$E[X] = 0 + 8.5 + 7.2 = 15.7$$

2. variance:

$$Var(X) = 133.10 + 29.41 + 235.50 = 398.01$$

3. standard deviation:

$$SD(X) = \sqrt{398.01} = 19.95$$

b. Assume independence between individual fliers (a rough but helpful approximation).

$$E[\sum_{i=1}^{120} X_i] = \sum_{i=1}^{120} E[X_i] = 120 \times 15.7 = \$1,884$$

$$Var[\sum_{i=1}^{120} X_i] = \sum_{i=1}^{120} Var[X_i] = 120 \times 398.01 = 47,761.20$$

$$SD[\sum_{i=1}^{120} X_i] = \sqrt{47,761.20} = 218.54$$

Question 3.44

a.

$$E[X + Y_1 + Y_2 + Y_3] = E[X] + 3 \times E[Y] = 48 + 3 \times 2 = 54$$

$$Var[X + Y_1 + Y_2 + Y_3] = Var[X] + 3 \times Var[Y] = 1 + 3 \times 0.0625 = 1.1875$$

$$SD[X + Y_1 + Y_2 + Y_3] = \sqrt{1.1875} = 1.09$$

b.

$$E[X - Y] = E[X] - E[Y] = 48 - 2 = 46$$

$$Var[X - Y] = Var[X] + Var[Y] = 1 + 0.0625 = 1.0625$$

$$SD(X - Y) = \sqrt{1.0625} = 1.03$$

c. Initially we do not know exactly how much ice cream is in the box. Then we scoop out an unknown amount. We should now be even more unsure about the amount of ice cream that is left in the box

Question 4.4

a. Let X denote the finishing times of Men (ages 30–34) and Y denote the finishing times of Women (ages 25–29):

$$X \sim N(\mu = 4313, \sigma = 583)$$
$$Y \sim N(\mu = 5261, \sigma = 807)$$

b. The Z-scores can be calculated as follows:

$$Z_{Leo} = \frac{4948 - 4313}{583} = 1.09$$
$$Z_{Mary} = \frac{5513 - 5261}{807} = 0.31$$

- Leo finished 1.09 standard deviations above the mean of his group
- Mary finished 0.31 standard deviations above the mean of hers

c. Mary ranked better since her Z-score indicates a relatively shorter finishing time compared to her group

d. Probability for Leo:

$$P(Z > 1.09) = 1 - P(Z \leq 1.09) = 1 - 0.8621 = 0.1379$$

So, **13.79%** of men in Leo's age group performed worse

e. Probability for Mary:

$$P(Z > 0.31) = 1 - P(Z \leq 0.31) = 1 - 0.6217 = 0.3783$$

So, **37.83%** of women in Mary's age group performed worse

f.

- The answer to part (b) would not change since Z-scores can be calculated for non-normal distributions.
- However, parts (c) through (e) rely on the **normal model** and cannot be answered without it.

Question 4.6

a. The fastest 5% are in the 5th percentile of the distribution. The Z-score corresponding to the 5th percentile of a normal distribution is approximately `-1.65`:

$$Z = -1.65 = \frac{x - 4313}{583}$$
$$x = -1.65 \times 583 + 4313 = 3351 \text{ seconds}$$

The fastest 5% of males in this age group finished in **less than 56 minutes**

b. The slowest 10% are in the 90th percentile of the distribution. The Z-score corresponding to the 90th percentile is approximately `1.28`:

$$Z = 1.28 = \frac{y - 5261}{807}$$
$$y = 1.28 \times 807 + 5261 = 6294 \text{ seconds}$$

Question 4.24

Let $X \sim Binomial(n = 3, p = 0.25)$ and for part (d) let
$X \sim Geometric(p = 0.25)$ (counting the trial of the first success)

a.

$$P(X = 2) = \binom{3}{2}(0.25)^2(0.75) = 0.1406$$

b.

$$P(X = 0) = \binom{3}{0}(0.25)^0(0.75)^3 = 0.4219$$

c.

$$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.4219 = 0.5781$$

d. first success on trial 3 (Geometric)

$$P(X = 3) = (1 - p)^{3-1}p = (0.75)^2(0.25) = 0.1406$$

Question 4.28

a. Negative Binomial Distribution with $n = 15$ and $p = 0.65$

$$P(X = 10) = \binom{14}{9}(0.65)^{10}(0.35)^5 = 0.1416$$

b. Use Binomial Distribution

$$P(X = 10) = \binom{15}{10}(0.65)^{10}(0.35)^5 = 0.2123$$

c. Use Geometric Distribution

$$P(X = 3) = (0.35)^2(0.65) = 0.0796$$